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Weak Partitioning Chromatography for Anion Exchange Purification of Monoclonal Antibodies

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ABSTRACT: Weak partitioning chromatography (WPC) is an isocratic chromatographic protein separation method performed under mobile phase conditions where a significant amount of the product protein binds to the resin, well in excess of typical flowthrough operations. The more stringent load and wash conditions lead to improved removal of more tightly binding impurities, although at the cost of a reduction in step yield. The step yield can be restored by extending the column load and incorporating a short wash at the end of the load stage. The use of WPC with anion exchange resins enables a two-column cGMP purification platform to be used for many different mAbs. The operating window for WPC can be easily established using high throughput batch-binding screens. Under conditions that favor very strong product binding, competitive effects from product binding can give rise to a reduction in column loading capacity. Robust performance of WPC anion exchange chromatography has been demonstrated in multiple cGMP mAb purification processes. Excellent clearance of host cell proteins, leached Protein A, DNA, high molecular weight species, and model virus has been achieved.

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KEYWORDS: weak partitioning chromatography; anion exchange chromatography; monoclonal antibodies; protein purification

Introduction

The surge in the number of therapeutic monoclonal antibodies (mAbs) in clinical evaluation has driven companies to develop platform manufacturing schemes flexible enough to accommodate most if not all of their future pipeline mAbs. The benefits of these platform processes are clear (Shukla et al., 2007), and extensive optimization has been undertaken in many process development groups to refine their mAb purification platform for improved product yield and purity, operational efficiency, and robust operation.

A typical platform for cGMP purification of mAbs derived from recombinant cell culture employs three chromatographic steps, with a Protein A (ProA) capture column followed by two chromatographic polishing steps (Fahrner et al., 2001; Shukla et al., 2007). The polishing steps almost invariably employ anion exchange (AEX) operated in a flowthrough mode, and one additional adsorptive step (either cation exchange, ceramic hydroxyapatite, or hydrophobic interaction). The polishing steps reduce levels of host cell protein, leached ProA, DNA, and undesired product variants such as high molecular weight (HMW) species. In addition, the polishing steps contribute to the viral safety of the product, as determined by virus clearance studies using model viruses. Sufficient clearance of trace impurities requires a reduction of several logs from the levels in the ProA pool, and the removal of HMW species can be challenging if their levels exceed a few percent of the product mass.

The next significant advance in mAb purification platforms will be the elimination of one of the two polishing steps, enabling the use of a two-column process. This will have significant advantages for cost of goods, facility fit, and operational ease, as well as process development and validation costs. AEX chromatography performed in a flowthrough mode removes acidic impurities (Gagnon, 1996), and is the logical choice from which to develop a single superior processing step which can replace two polishing steps.

A re-evaluation of AEX chromatography for mAb purification has led to the establishment of weak partitioning chromatography (WPC), which is distinct from bind-elute and flowthrough operations. In contrast to AEX flowthrough, the superior performance of AEX in the WPC mode enables a two-column platform. This article

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introduces the concept and operating principles of WPC, describes methods of defining WPC operating windows for AEX purification of mAbs, and discusses the influence of product binding on step recovery and column loading capacity. Summary performance data on impurity clearance and product yield for multiple cGMP mAb purification processes are also presented.

Materials and Methods

Antibodies

The monoclonal antibodies (mAbs) used for these studies were murine, chimeric, or humanized IgG of various isotypes produced in recombinant Chinese hamster ovary (CHO) cells grown in a serum-free medium. Partially purified preparations which had undergone capture by ProA chromatography resins were used; these materials had representative levels of host cell protein and residual ProA which had leached from the affinity capture resin.

High Throughput Screening (HTS)

A detailed description of the configuration and use of the robotic liquid handling unit used in the high throughput screening (HTS) can be found in previous publications (Coffman et al., 2008; Kelley et al., 2008). A Tecan Genesis 150 (Tecan US, Research Triangle Park, NC) was used for liquid and resin handling. A 96-well filterplate (Innovative Microplate, Billerica, MA p/n F20022), with a 0.45 μm PVDF membrane was used to incubate resin with the protein and buffer. The filterplate was centrifuged at 1,200*g* to separate the solution from the resin. The filtrate solution was captured in a collection plate and then analyzed using a Spectramax Plus 96-well UV–Vis plate reader (Sunnyvale, CA). All experiments were performed at room temperature. The protein stock was typically concentrated to 10 or 40 mg/mL (depending on the resin challenge) and dialyzed into 1 mM HEPES, 10 mM NaCl buffer. Fifty microliters of AEX resin was added to each well, which was then equilibrated to the appropriate pH and sodium chloride concentration. The protein load challenge was 5 mg/mL of settled resin. After mixing the equilibrated resin and protein solution for 20 min on an orbital shaker, the unbound protein in the supernatant was collected and analyzed.

Anion-Exchange Chromatography Operating Conditions

Fractogel® EMD TMAE HiCap (M) anion exchange resin was obtained from EMD Merck (Darmstadt, Germany). All laboratory chromatographic experiments were carried out using an AKTA FPLC chromatographic system from GE Healthcare (Uppsala, Sweden). Laboratory columns ranged from 0.5 to 0.66 cm in diameter and were 10–20 cm in height. The columns were equilibrated to the specified operating conditions of pH and total chloride content prior to loading. ProA peak pools were then applied to the column followed by a 3 column volume (CV) wash of the equilibration buffer. The load eluate and wash volumes were collected as the process pool. The remaining bound protein was removed using a 2 M NaCl strip buffer. The columns were sanitized with 2 M NaCl in 0.5 M NaOH and stored in 16% ethanol, 150 mM NaCl in 20 mM Tris, pH 7.2. cGMP anion exchange chromatography was performed with 40–60 cm diameter columns, with bed heights of 11–25 cm. Load challenges at cGMP scale varied from 18 to 200 g protein per liter of resin.

Analysis

Protein concentrations were determined using UV absorbance at 280 nm. Analytical scale high performance size exclusion chromatography (HPLC-SEC) was used to measure HMW content. The concentration of CHO cell proteins was measured with an immunoassay employing BioVeris technology. Residual ProA was measured using an ELISA. Residual DNA was measured using an in-house qPCR assay.

Distribution coefficients for the resin–protein interaction derived from the HTS experiments were fit using a response surface model containing the log chloride concentration and pH as main factors, plus their two-way interaction, and their two quadratic terms, as described previously (Kelley et al., 2008). The R^2 values for these models are typically >0.9, and residuals consistently fall within 0.2 of the measured log of the distribution coefficient.

Virus Clearance Studies

Virus clearance studies were conducted on scale-down models of the cGMP process steps following the well-established rules of chromatography scaling (Sofer, 1996). The lab-scale systems were qualified as suitable models of the full-scale cGMP step by evaluation of chromatograms, product yield, and purity (Godavarti et al., 2005). The virus titers were determined using plaque- or foci-forming in vitro cell-based assays. Statistical methods were employed to convert the viral titers of the test samples into concentrations.

Background

Definition of Weak Partitioning Chromatography, Operating Concepts, and Product Recovery Considerations

An isocratic mode of chromatography named weak partitioning chromatography (WPC) is described here in quantitative terms. To make a distinction in the chromatographic nomenclature, the term "weak partitioning" is meant to describe the distinct zone of binding strength of the product to the resin during this isocratic operation. It is not meant to refer specifically to any one type of liquid chromatography, such as "partition" chromatography, a term that is used when exploiting relative solubilities in the stationary and mobile phases. The AEX polishing of partially purified monoclonal antibody (mAb) pools is used to illustrate the principles and benchmark the performance of WPC.

The chromatographic separation of a protein from impurities requires that these species exhibit differential interactions with the chromatography resin. The distribution coefficient (sometimes called the partition coefficient), which is the ratio of the concentration of the solute bound to the resin divided by the concentration in solution at equilibrium, can be used to describe the strength of this interaction. Equation (1) defines this constant for a given protein (or other impurity) as K_p.

$$K_p = \frac{Q}{C} = \frac{[\text{bound}]}{[\text{free}]} \tag{1}$$

We have measured K_p for proteins at low concentrations of bound protein (<5 g/L), to ensure that the distribution coefficient also represents the initial slope of the adsorption isotherm. K_p is used in this article instead of k', the capacity factor, because most of the data used to characterize the resin–protein interaction come from batch-binding experiments, where a distribution coefficient is more commonly used than k'. K_p can be converted to k' using the phase ratio of a packed chromatographic column (Velayudhan and Horvath, 1986). For ion exchange chromatography, the strength of interaction with the resin is modulated by solution pH and the concentration of counterions in the mobile phase. The protein–resin interaction can often be described by a power law function relating the distribution coefficient at a given pH to the counterion concentration (Boardman and Partridge, 1955; Kopaciewicz et al., 1983; Velayudhan and Horvath, 1986). Figure 1 is a schematic diagram showing this log-log dependence of product and impurity K_p values on the counterion concentration for an AEX resin.

Protein ion exchange chromatography separations are typically run in one of two modes. In the bind and elute (B/E) mode, the protein product is bound tightly under loading conditions (the product K_p is very high, often >100); the column is then washed to remove loosely bound impurities, and the protein is eluted by a change in mobile phase pH or counterion concentration (the product K_p during elution is very low, often <0.1). When AEX chromatography is used for mAb purification, it is more common to run in a second mode known as an isocratic flowthrough (FT) mode. Under these conditions, the product does not bind to an appreciable extent to the resin ($K_p < 0.1$), and the product is collected in the load eluate, also called the column flowthrough. When designing a B/E or FT ion-exchange step, the K_p values for load and elution

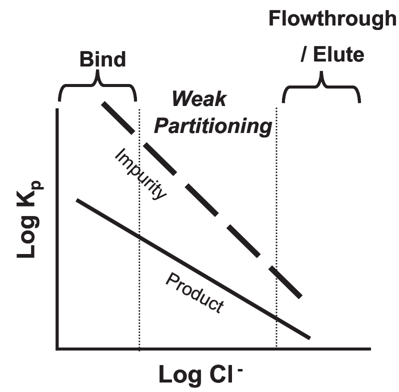

Figure 1. Schematic diagram of anion exchange chromatography, showing the log of the distribution coefficient, K_p, for both the product mAb and an impurity as a function of the log of the counterion concentration.

conditions modes are not often used; instead, measurements of product losses during B/E load and wash steps and the recovery in the product pools for B/E and FT modes are used for optimization. Adjustments to the solution composition are made if the product losses are unacceptable (more stringent binding and/or stronger elution conditions). If the losses are minimal but the impurities are not sufficiently removed, less stringent binding and weaker elution conditions may be tried. After optimizing operating conditions for acceptable product recovery and impurity clearance, the impurity level is assessed from an optimized run and compared against the target.

The hypothetical example in Figure 1 shows the case where an impurity binds to the resin more tightly than the product. This is typical for AEX chromatography of most neutral or basic proteins and impurities such as nucleic acids, endotoxin, virus, and often host cell proteins, which are generally acidic for both mammalian and *E. coli* cell lines (Champion et al., 2003; Van Dyk et al., 2003). For this reason, AEX polishing steps are often run in FT mode (Fahrner et al., 2001). The major benefit of a FT operation compared to B/E is that the resin capacity is used to bind only the impurities and not the product, which is the major constituent of the load. Consequently, the relatively large process volumes may be applied with the ease and simplicity of an isocratic, volume overloaded operation (Gareil et al., 1980; Knox and Pyper, 1986).

Optimization of an AEX step in the FT mode by reducing the counterion concentration of the column load at a constant pH generally gives rise to increased product losses, since the product binds more strongly to the resin. A stepwise optimization in which the counterion concentration is decreased at a constant product loading (typically 50–100 g of product per L of resin) will eventually give rise to unacceptable product losses. The lowest counterion concentration in the load that would result in acceptable product yield is chosen. With this approach, the column operation would be considered to have been optimized for product recovery and impurity clearance for this pH under FT conditions.

For a WPC operation, which is also an isocratic operation, the optimization algorithm responds differently to the reduced yield at the lowest counterion concentration; the counterion concentration of the load could even be reduced further, but in concert with loading more product onto the column in order to raise the product yield back to acceptable levels. While simulations of product recovery during isocratic AEX operation would be more informative and accurate, consider the following simplified situation: a 20% product loss at 50 g/L column loading means that the product remaining bound to the resin following collection of the product pool is 10 g/L, a very high amount of product lost in the column strip. However, at higher loadings, the mass of product bound to the resin would be unchanged (assuming that the load volumes for 50 g/L and higher loadings will lead to an equilibrium concentration of product bound to the resin; this is often the case for load volumes of more than 2–3 column volumes (CVs) under FT conditions). If the column was loaded to 250 g/L instead of 50 g/L, this 10 g/L loss in the column strip would only account for 4% of the load, which is a much more acceptable loss. Now, if a brief isocratic wash ($\sim$3 CVs) is included in the product pool after the load is completed, much of the product bound to the resin can be recovered. This would enable even lower counterion concentrations to be used in the load and wash, and lead to even higher concentrations of product binding the resin. The benefit of this continued decrease in counterion concentration is an increase in the binding strength of the impurities, which results in a greater reduction in the impurity level in the product pool. As shown in Figure 1, the K_p of the impurity is also increasing as the counterion concentration drops. Hence, WPC defines a third mode of chromatography using isocratic conditions wherein a significant amount of product binds, thus allowing enhanced binding of impurities relative to the FT mode. Product losses are reduced by loading to high levels and employing a short wash step, so that both better purity and a higher loading capacity are realized.

WPC is defined under conditions where the product K_p is between typical B/E and FT modes. The K_p values in the WPC mode range from 0.1 to 20, most preferably between 1 and 3 for AEX examples in this article. The equilibrium product binding to the resin typically ranges between 1 and 20 g/L. With product binding at levels up to 20 g/L, WPC operation is distinct from FT. The product load challenge in the WPC mode is often 250 g/L, but could be much higher for some applications (data to follow). The short column wash recovers most of the bound product; if the impurity K_p is greater than 50–100, then there will be a minimal decrease in the product pool purity during the wash. Further, the product pool is not diluted significantly by the wash, as the load volumes are often 15–30 CVs (corresponding to just 20% and 10% dilutions of the product pool, respectively).

Nature of Impurities and Enhanced Removal by Weak Partitioning Chromatography

The K_p values for various impurities can be measured (data to follow), and the selectivity (α) is calculated using Equation (2):

$$\alpha = \frac{K_{P,\text{impurity}}}{K_{P,\text{product}}} \tag{2}$$

In general, the more stringent loading conditions used in WPC are intended to increase the impurity K_p. Both the $\log_{10}$ reduction values (LRVs) of the impurity removal and the loading volume at which point the impurity breaks through can be significantly increased. If the impurity K_p value is on the order of the load CVs, the impurities will bleed through in the load eluent or wash. For a given binary separation, α might increase with product K_p (as it does in Fig. 1), yet higher product pool purity can be achieved with higher impurity K_p even if α is constant or decreases slightly. The interpretation of an impurity K_p and its connection to the LRV provided by the column step will depend somewhat on the characteristics of the impurity itself. Different impurities in a mAb feedstream vary in terms of heterogeneity and their interaction or relation to the product, and so it is worth discussing them here.

Host cell proteins are a heterogeneous impurity of many different protein species produced by the host cell. Most CHO proteins are acidic, yet these molecules vary in size, charge, and hydrophobicity. The K_p measured for a HCP mixture represents a weighted average of many HCP species that exhibit strong to weak interactions with the chromatographic resin. As a consequence, isocratic chromatographic steps conducted under higher K_p values for HCP (and product) will result in increasingly large LRVs in the initial column effluent fractions, as will be illustrated later. This result is not necessarily an expected outcome for an ideal binary separation, where an increasingly large impurity K_p would not translate to a better purity in initial effluent fractions, although it would allow for larger load volumes. The advantage of increasing an average K_p for a heterogeneous impurity is that the K_p's of the weakest binding species are increased enough for them to be more effectively removed.

DNA, viruses, and endotoxin are heterogeneous impurities which need to be removed by this purification step. However, optimization of DNA or virus removal does not appear very sensitive to WPC conditions examined (data not shown). For example, DNA fragments vary in size, but should always be negatively charged and bind tightly to an AEX resin at moderate or low counterion concentrations (certainly, the LRV's for these impurities will suffer at higher counterion concentrations). Likewise, virus clearance at the low ionic strengths used for AEX-WPC should be both significant and robust.

When considering the impurity behavior from various mAb feedstreams, HCP, DNA, endotoxin and viruses are all classified as being non-product related. Barring any non-specific or specific interactions with the product itself, they should behave independently of the product. Consequently, the impurity LRV's should be similar for WPC-AEX steps operated in the same salt and pH conditions for different mAbs, regardless of differences in the mAb K_p values.

The leached ProA species in a ProA product pool will also be heterogeneous, arising from proteolysis of the ProA ligand at multiple sites (Carter-Franklin et al., 2007). In contrast to the impurities described above, leached ProA presumably exists in the feedstream as a ProA-mAb complex. As a result, ProA can be considered "product related," and the ProA K_p will depend on the product K_p, generally exhibiting parallel (but more tightly bound) K_p contours, unlike HCP.

Product related variants, such as aggregated or high molecular weight (HMW) species, might also bind more tightly than the product. AEX often shows promise in reducing HMW, probably through the greater avidity of the HMW species for the AEX resin; with the larger surface area of the HMW species having more potential sites of attachment, the HMW would likely bind tighter to the resin, and therefore be amenable to removal by WPC or FT chromatography. Generalizations on HMW removal using AEX are not straightforward. The characteristics of HMW species are product specific, even for a family of humanized mAbs. For cases where HMW removal by AEX is possible, a HMW K_p can be determined (for each species, if resolvable peaks are observed using SEC-HPLC), and the general observation is that α increases slightly with increasing monomer K_p. In some cases, the HMW species do not bind more strongly on AEX despite the potential for increased avidity by a larger species (the HMW species are acting as more basic, not more acidic, species). For these loads, a different resin from AEX must be considered for HMW removal, as well as AEX in the B/E mode, where the HMW could potentially be removed in the load and wash stages.

The general trend observed in WPC is an increasing LRV of an impurity as the product K_p (and more meaningfully, the impurity K_p) is increased. However, there is a limit to the benefit of increasing the product K_p during the load and wash. When very high product K_p values are reached, the product will begin to occupy a significant fraction of the total resin capacity and can cause an early break through of impurities. The influence of product K_p on the column loading capacity can be complex, as the level of impurities in the load will also affect the column capacity. This will be illustrated for cases where a significant quantity of a mAb product is aggregated, and binds more tightly than the monomer. The mass of HMW can be orders of magnitude greater than other proteinaceous impurities such as HCP or leached ProA. In the case where the HMW level is >5% of the load mass, the column breakthrough can occur well below the 250 g/L value typical for the load of a WPC step having only trace levels of protein impurities. Hence an optimum product K_p exists for each separation, and will depend upon the process step objectives for product recovery, impurity clearance, and target column loading capacity (this will be shown later with experimental data).

WPC should be amenable to some of the standard chromatographic column modeling approaches commonly applied to process chromatography (Ishihara et al., 2007; Iyer et al., 1999; Karlsson et al., 2004). Simplistic modeling of product recovery from column runs based solely on the product isotherm has been useful to predict trends as function of K_p, load challenge, and product concentration in the load (data not shown). Accurate predictions, however, would need to take into account the complexities of multi-component equilibria with the resin, and the competitive effects of product as well as major impurities.

In summary, WPC uses more stringent binding conditions than FT during the load and wash stages, often with only modest changes in counterion concentration or pH. This causes an increase in the product K_p, which results in stronger binding of product and impurities. The column loading is increased, and a short wash is included to maximize the product yield. Operationally, a WPC step uses the same stages of operation as an isocratic FT step, but enables higher loading and results in better product purity.

Results and Discussion

Mapping Potential WPC Conditions Using HTS

The first step in exploring WPC is to map the product and impurity interactions with the resin as a function of the mobile phase variables affecting binding strength. Two methods will be described: first, HTS using batch-binding, and then a series of column runs with decreasing counterion concentration. This section will cover the case where the AEX column load has only trace impurities (<5% by mass). The more complicated case of dealing with less pure load material will be addressed in a separate section below.

When evaluating batch-binding with HTS systems, the product and impurity binding on AEX are determined as a function of pH and counterion (anion) concentration (Coffman et al., 2008; Kelley et al., 2008). A response surface of these two variables is programmed into a robotics liquid handling system, and a resin slurry is dispensed into a 96-well filterplate. After equilibration of the resin at a unique combination of pH and counterion concentration in each well, a protein solution of the same buffer composition is added to the resin, the slurry is agitated to establish equilibrium, and then centrifuged to separate the supernatant from the resin. The protein and impurity concentrations are measured in the filtrate, and used to calculate the K_p of the product and impurities. The amount of product offered to the resin is kept low (<5 g/L of resin), to ensure that the protein challenge is in the linear portion of the equilibrium binding isotherm. Note that these procedures of HTS can be applied to other chromatographic modes (Kramarczyk et al., 2008; Wensel et al., 2008), and provide similar information for other resins.

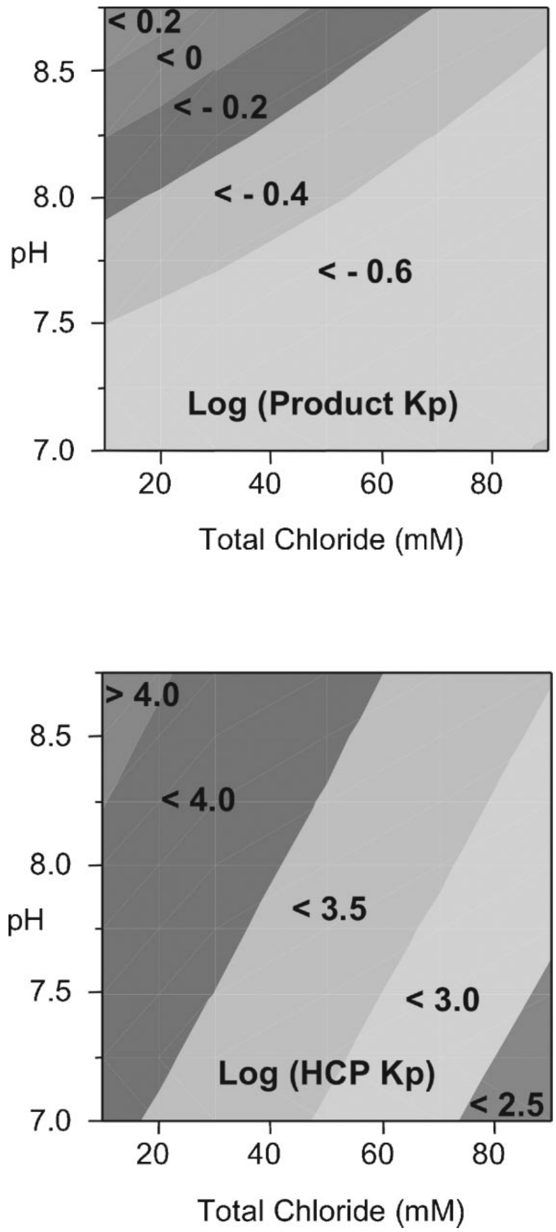

Figure 2. Results of high throughput screening (HTS) of an AEX resin under batch binding conditions. Contours of constant values of K_p for a mAb (product) and an impurity (host cell protein, HCP) are shown (the K_p contours were generated using a response surface model derived from the raw data).

Figure 2 shows the product K_p as a function of pH and counterion concentration for a monoclonal IgG. A response surface model was fit to the data using analysis of variance, and subsequently used to estimate the K_p for any combination of pH and counterion concentration in the experimental space. The response surface models fit the AEX datasets for various mAbs very well, and can be expected to give estimates of K_p within 10% of the actual value. The response surface for the HCP K_p for this feedstream is also shown in Figure 2. In this case, the load to the HTS system was a partially purified ProA pool containing several thousand ppm of HCP.

Another way to interpret impurity and product K_p values is shown in Figure 3, which is referred to as a selectivity plot. Here, the impurity K_p (leached ProA in this case) is plotted against the product K_p on a log-log plot. The tight correlation between the two distribution coefficients is commonly observed with IEX chromatography operated in the WPC mode for product-related impurities. This tight correlation suggests the separation at a given K_p is independent of pH and counterion concentration, and the basis of separation is constant throughout the explored operating space. Note that the same plot for HCP would show more scatter, since the product and HCP K_p contours shown in Figure 2 are not parallel. It is worth noting that the slope of the plot is greater than one, indicating that α increases with product K_p.

The window of operation for the product K_p which will give rise to successful WPC separations can be determined for any resin–product pair by further exploration through column experiments (one approach is described below). The upper limit of product K_p is dictated by a minimum target for the product recovery in the load eluate and wash stages. The lower limit of the product K_p would be determined by the minimum impurity LRV required of the separation. Once experience is gained in defining the K_p range for a specific resin, one can quickly establish an operating window of product K_p that meets the step objectives. Figure 3 shows a window of operation for the product K_p of 1–3, which is preferable for AEX chromatography using Fractogel TMAE HiCap for mAb polishing (in some circumstances, a K_p of over 10 may be preferable). The response surface model of the product K_p as a function of pH and counterion concentration derived from the HTS experiment (Fig. 2) can

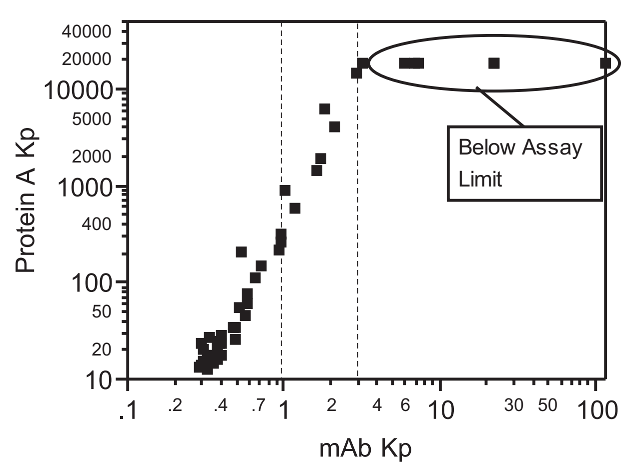

Figure 3. The relationship between the K_p of leached ProA residual and the mAb K_p for the TMAE Fractogel HiCap AEX resin. This selectivity plot was generated using a high throughput screen, and is useful in indicating the potential of the separation to remove impurities to target levels. At high values of product K_p, the ProA ELISA test values are below the limit of quantitation of the assay. The dashed lines indicate the preferred window of operation for mAb purification using this AEX resin ($1 < K_p < 3$).

then be used to define the combination of pH and counterion concentrations that are allowed within the product K_p window of operation. The counterion concentration range covering K_p from 1 to 3 depends on the pH and antibody, but typically is a 10–20 mM window for chloride systems, with midpoint counterion concentrations ranging from 20 to 60 mM total chloride.

Operating anywhere along a contour of constant product K_p should result in similar product yields and chromatograms, if all other variables such as load challenge, product concentration and wash volumes are held constant. This is based on the observation from HTS studies that the product binding at high resin challenges is directly correlated to the log of the product K_p (Kelley et al., 2008), suggesting that a protein's adsorption isotherm will be similar at a given K_p. However, the selectivity of the separation could differ along this contour of constant product K_p, so it is useful to explore the full range of pH and other mobile phase modulators allowed. If the selectivity plot from the HTS evaluation shows some scatter and not a single universal curve as seen in Figure 3, then differences in mobile phase composition may offer improved selectivity. The impurity LRV that is observed during column operation may be sensitive to modest changes in pH and counterion concentration. Changes that give rise to a slight increase in product K_p (two- to threefold) can lead to significant improvements in impurity removal. This should be tested with column runs to determine if the difference is real and whether it has any process significance.

The product K_p alone does not determine product yield. As the product K_p increases, yield becomes sensitive to the load challenge. As described above, operating at a high product K_p and too low of a load challenge could result in an unacceptably low yield. The maximum allowed load challenge is dictated by the impurity breakthrough in a column capacity study. For situations where there are only trace levels of impurities, the column loading capacities for product can be very high (over 1,500 g/L in some cases). Because extremely high loadings cause long processing times when very small columns are used at their maximum capacities, a compromise may be struck with regards to operating at high loadings.

In the absence of HTS data, column runs can easily be used to explore potential WPC conditions. Beginning with FT conditions at moderate counterion concentrations (70–100 mM total chloride for many mAb/AEX resin pairs), a series of column runs can be conducted with reduced counterion concentrations at constant pH values; the columns should be run at product challenges of 100 g/L or higher. One approach is to use a geometric series, in which the next run's counterion concentration is a specified percentage of the previous run, which gives rise to constant intervals of the log of the counterion concentration. As the counterion concentration drops, the chromatogram will eventually indicate the WPC region where significant product binding begins to take place, as explained below. The upper limit of the product K_p will be evident when the product recovery is below the acceptable limit even after a modest column wash (the equilibration and wash solution should be the same, and match the counterion concentration of the protein load). Fine-tuning of the counterion concentration in this region will help determine the optimal mobile phase composition of the load and wash stages. This iteration can be repeated at other pH values, if it is suspected that the selectivity for an impurity could vary with pH.

Column Experiments: Assessing Recovery, Impurity LRV, and Breakthrough Profiles

The initial column experiments used for exploring AEX WPC for mAb polishing should be conducted with product K_p in the range of 0.1–20, preferably at a product K_p of 1–3 for an AEX step using TMAE Fractogel HiCap, the resin used in these experiments. At a low load challenge of approximately 100 g/L, the impurity removal typically improves with increasing K_p. However, competitive effects lead to a decrease in loading capacity, which is only revealed in a capacity breakthrough study; these studies are typically done after the scouting runs at low challenges in order to preserve load material. The studies described in this section focus on removal of HCP and leached ProA, but the approach will be similar for optimization of removal of other trace impurities.

Figure 4 shows an overlay of two runs with product K_p values of <0.1 and 2. The shift in the product breakthrough during the load is easily detected, and indicates WPC operation. This shift in the breakthrough curve arises from the increased product mass binding to the resin at the higher product K_p. The strip peak area is increased in WPC, due to both the increase in impurities retained by the column, and from the additional product not completely washed off the column. It is valuable to carefully analyze the column strip fraction, as this is the most sensitive indicator of product

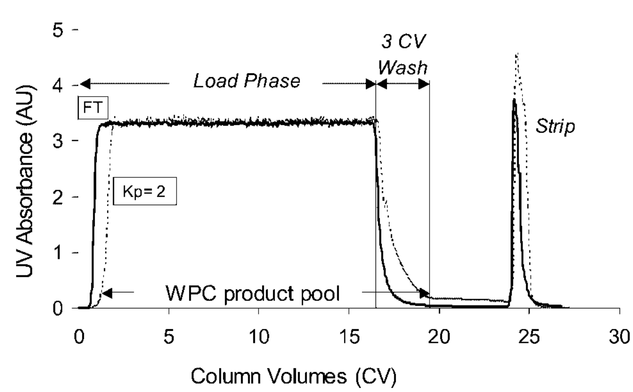

Figure 4. Chromatograms from column runs under flowthrough ($K_p < 0.1$) and weak partitioning conditions ($K_p = 2$) display the effects of increasing mAb K_p on the product breakthrough and washout regions of the chromatogram. The column strip peak is also larger, due to the increased removal of host cell proteins and a slight increase in product loss.

loss; a 5% product loss is more accurately determined by analysis of the column strip than from measurements of the recovery in the product pool, typically. This is due to the lower total mass in the strip fraction, and the improvement in signal-to-noise ratio of the measurement (because there may be a significant mass of impurities in the column strip fraction, a product specific assay such as ProA-HPLC may be needed to determine the mass of product loss).

Figure 5 shows the HCP LRV for challenges as high as 1,500 g/L for a series of four AEX column runs, and illustrates the benefits as well as the competitive effects that can be observed by increasing product K_p. The lowest product K_p tested, <0.1, corresponds to FT conditions. Note that for this condition, the HCP LRV declines from 2 to 1 in the early column effluent fractions, as the more weakly bound HCP species bleed through the column during the loading. Initially, increasing the product K_p into the WPC region increased the effluent purity, as the HCP bound more tightly, providing a significant improvement in HCP LRV and loading capacity. At a load challenge to 250 g/L, the HCP LRV has risen from 1.5 for FT conditions to almost 4 for the highest product K_p tested, which was $K_p = 7$. This increase in HCP LRV has important process significance, as 4 LRV of HCP removal is typically sufficient to reduce HCP levels in a ProA pool to target values for final product, thus enabling a two-column process.

The eventual impact of increasing product K_p on column loading capacity is also evident in Figure 5. The column loading at which point HCP breaks through declines significantly as a high degree of product binding competes for capacity on the resin at product K_p values above 2.7. The capacity is roughly halved as the product K_p is increased from 2.7 to 7. At a K_p of 7, approximately 10–15 g/L of mAb is binding to the resin at equilibrium, depending on the concentration of product in the load.

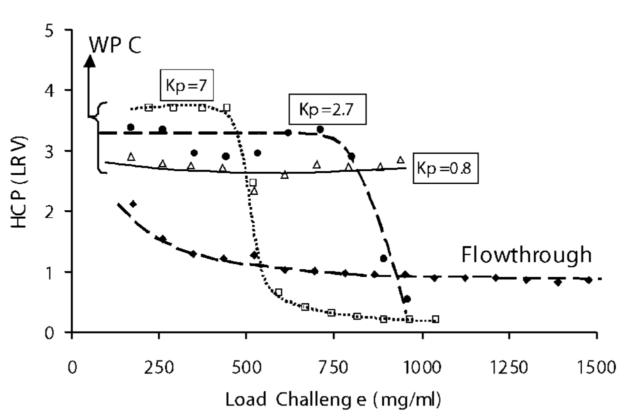

Figure 5. The log reduction value (LRV) of host cell protein and the column loading capacity at breakthrough depend on the mAb K_p. Under flowthrough conditions with low product K_p values, the LRV is modest and declines due to the weak binding of impurities. In the weak partitioning region, the LRVs increase with product K_p and the capacity may be extended. At higher product K_p, the product binding takes up a significant amount of resin's binding sites, and the column capacity is reduced.

The minimum acceptable LRV and impurity breakthrough capacity together define the operational product K_p range for a given separation. For this process, a K_p of 0.8 was chosen, and the loading capacity set to 250 g/L. This load challenge is a conservative limit, and should allow for robust product recovery and impurity clearance. (A process characterization study will be described below that evaluates process control parameters of pH and counterion concentration to more effectively prove this point.)

It is intriguing to consider the process implications of the case where the column loading capacity exceeds 1,000 g/L. In membrane chromatography, there are clear benefits of very high device loadings (Zhou and Tressel, 2006), however, some of these advantages are lost when chromatographic columns can also be run at very high load challenges. At some point, the column productivity (g/L/h) suffers if excessive load challenges are combined with a small column volume. Operating a column with a short bed height and increased cross-sectional area would be one approach to maximizing productivity, while still enjoying the benefit of elevated load challenges.

The final process operating conditions can be tested after the loading capacity runs at the optimal conditions for product recovery and impurity removal. This would comprise a WPC run using these final conditions, but with the column challenged to only 50–80% of the full capacity, and incorporating a short column wash after the load. The product recovery and the LRVs for all impurities are determined under these optimized conditions. Table I compares the results from an optimized WPC step with a FT step for processing a common feedstream (a ProA pool for a humanized monoclonal IgG with only trace levels of impurities). Table II summarizes the analytical data on the load, product pool, and strip for a study using a different mAb feedstream. Although an exact mass balance for impurities such as HCP and ProA is difficult to obtain using an ELISA (>50% errors in the mass balance are seen here), analyses of the strip fraction supports that the additional clearance of impurities in the product pool arise from their stronger binding to the resin. The additional product loss in the strip, in terms of mass of product per volume of resin, is compensated by the increased loading to the column and results in an acceptable recovery (increased loading for the FT condition would result in further decreases in impurity LRV, as observed in Fig. 5). As discussed above, analysis of the strip in this dataset provided a more accurate measure of product loss than the product pool recovery calculation itself.

WPC of Less Pure Feedstreams With Elevated Levels of High Molecular Weight Species

For feedstreams with high concentrations of impurities (>5% by mass), the column loading capacity under WPC conditions may be reduced. There was no discussion of the removal of HMW species in the section on trace impurities

Table I. Comparison of AEX flowthrough and weak partitioning for polishing of a Protein A chromatography peak in a monoclonal antibody process.

Mode	Load challenge (g/L)	pH	Cl^- (mM)	HCP (LRV)	ProA (LRV)	Product recovery (%)	Product in strip (g/L resin)
FT ($K_p = 0.2$)	275	7.0	83	0.6	<0.1	>95	0.8
WPC ($K_p = 1.0$)	300	8.1	45	>2.6	>3.0	>95	5
WPC ($K_p = 5.2$)	300	8.0	17	>2.0	>2.6	>95	9

above; if HMW is a trace impurity, then all the principles of the previous section apply. For situations where the AEX column load has a large quantity of HMW, or some other impurity that will take up a significant amount of resin capacity, there are additional considerations.

Most three-column processes used for mAb purification do not rely on the AEX FT step to reduce HMW species to low levels. Usually, this is performed by a B/E chromatographic step using cation exchange, ceramic hydroxyapatite, or hydrophobic interaction. However, AEX-WPC is the sole HMW reduction step in the two-column platform process described in this article, and therefore must address the removal of HMW species.

Figure 6 shows HTS data on the selectivity for HMW separation from several mAbs which had up to 12% HMW in the load. Although the actual value for selectivity, α, is not calculated here, the decline in HMW percentage observed in the supernatant indicates an increase in selectivity with increasing product K_p. These monomer-HMW pairs show different selectivities and thus varied potential for AEX-WPC to remove HMW. For mAbs #1 and 2, the high selectivity made these separations relatively easy; for mAb #3, the process step was successful, but only after optimization of operating conditions and a reduction in column loading. Operating at higher K_p values becomes necessary for these challenging separations, which could cause lower product yields, or a reduction in total loading capacity due to elevated product binding. Note that the tight curves for mAbs #2 and #3 indicate that the ranges of pH and counterion concentration shown in Figure 2 did not influence selectivity at a given K_p.

Other patterns could also present themselves on selectivity plots. Although not observed for these HMW species, when an impurity does not bind the resin, the selectivity plot would be expected to show an increase in the relative impurity level of the batch-binding supernatant as K_p increases. This would occur because the concentration of product in the supernatant decreases significantly at high K_p values, while the concentration of impurity would remain constant, and hence the impurity content would increase on a relative basis. In this case, a B/E operation (instead of a FT or WPC operation) would likely be necessary for further optimization of the step for impurity removal.

When large quantities of HMW must be removed by a WPC or FT step, the column loading capacity will be reduced. Figure 7 shows a column breakthrough run for a mAb load with 18% HMW. The HMW breakthrough coincides with other impurities, and occurs at only 50 g/L, much lower than the column capacities in the trace impurity section above. While this reduction in column loading capacity is significant, a WPC or FT step will still have greater overall capacity for product compared to operating in the B/E mode. For this WPC step, the HMW was reduced to 1%, the product recovery was 90–95%, and the dilution factor was 1.5-fold. Figure 8 shows the LRV of HCP as a function of load challenge for the three mAbs shown in Figure 6, and highlights the impact that HMW levels have on column loading capacity. The rank ordering of the column capacities reflect the amount of HMW in the various loads. The total amount of HMW adsorbed to resin at the point of breakthrough for all of these mAbs is well below the maximum binding capacity of the resin for IgG monomer even if the HMW K_p is reasonably high. If the pores of the resin are too restrictive, steric exclusion or hindered diffusion of the large HMW species may limit the resin loading capacity (Yao and Lenhoff, 2006). Additionally, the conditions used in these examples, which are intended to maximize impurity binding (low conductivities and elevated pH), may lead to solute exclusion effects in which the tightly bound HMW species in the outer shell of the resin beads hinder further uptake of new impurities (Harinarayan et al., 2006). Resins with larger pore size distributions or enhanced surface diffusion mechanisms could potentially have larger loading capacities in WPC operations for these HMW containing loads.

WPC is still a useful chromatography mode for feedstreams with elevated HMW levels, and will typically offer the same performance with regards to clearance of HCP, leached ProA, virus, etc. as in the case of the more pure feedstream, albeit with a lower overall column loading capacity. Operating at a higher product K_p and lower load challenge could give rise to lower product recovery, and so

Table II. Composition of the column fractions from flowthrough and weak partitioning AEX steps.

	Load			Load eluent + wash			Column strip			
Mode (product K_p)	Challenge (g/L resin)	HCP (ppm)	ProA (ppm)	HCP (ppm)	ProA (ppm)	Product (% of load)	HCP (% of load)	ProA (% of load)	Product (g/L resin)	Product (% of load)
FT (0.1)	50	10,640	3	23	2	97	48	1	0.5	1.3
WPC (0.8)	250	10,640	3	6	<0.3	103	76	60	7.5	3.3

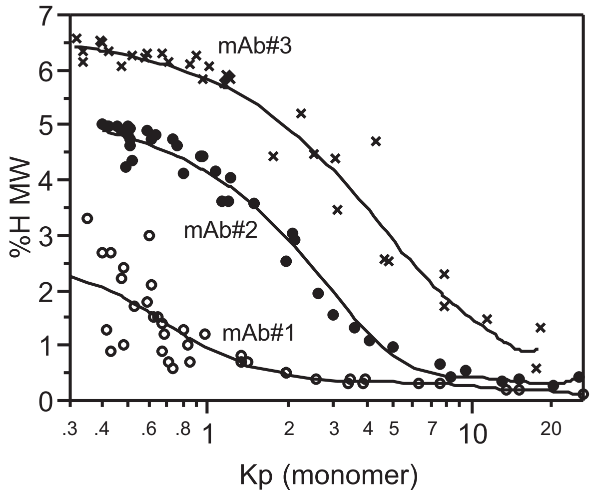

Figure 6. High molecular weight species (HMW) of mAb ProA pools can be removed by AEX chromatography, as revealed by HTS studies under pH and counter-ion concentration ranges shown in Figure 2. The selectivity plot shows excellent removal of HMW even at low mAb K_p values for mAb #1; for the other mAbs, the selectivity is not as good. For mAb#3, the HMW levels are not reduced to low levels until the product K_p approaches the upper limit of the weak partitioning chromatography range. The HMW content of the loads was 12% (mAb #1), 6.5% (mAb #2), and 6.5% (mAb #3).

the limits of the operation for worst-case performance for recovery should be evaluated.

Robustness of WPC Steps

As with any cGMP chromatographic step, the operation should be characterized to test how the extremes of both controlled and variable process parameters influence performance. Often fractional factorial or response surface studies are used for this purpose (Kelley, 2000; Seely, 2005). For WPC, the isocratic operation reduces the number of

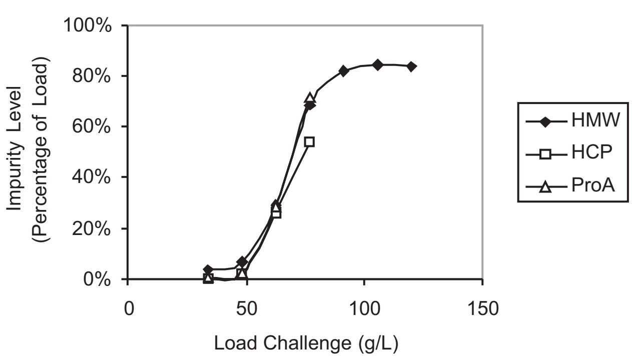

Figure 7. The impurity removal capacity of the AEX step is reduced when high levels of impurities such as HMW are retained. The resin saturates more quickly, and the tightly binding impurities breakthrough at roughly the same point.

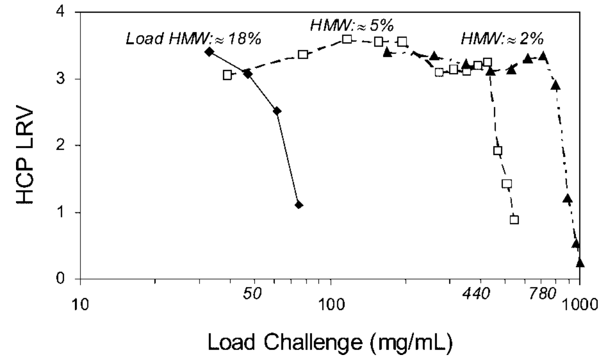

Figure 8. The HMW content of the load to the AEX step influences the column loading capacity for impurity removal. Higher HMW levels result in lower loading capacity in the column runs. The HCP impurity removal remains above 3 LRVs until the breakthrough of HCP, HMW, and other impurities is reached. The decline in HCP clearance when the column capacity is reached is rapid.

variables that need to be tested. Because both the product and impurity K_p are functions of pH and counterion concentration, these variables should be evaluated. Product recovery will vary with load challenge, and there will be normal process variations in the composition of the column feedstream. The lot-to-lot variability of the chromatographic resin could influence the separation as well.

To determine the robustness of an AEX-WPC step, a two-level full factorial study varying pH, counterion concentration, and resin ligand density (9 runs including a centerpoint run) was conducted at high load challenge. The study design and results are summarized in Table III. The pH was tested over 0.2 pH units, which is a normal control range for many chromatographic separations (the pH of the load is controlled by the titration of the ProA peak prior to loading onto the AEX column). The counterion concentration was varied ±20% from setpoint, which is a much wider range than would typically be observed in a well-designed process step. In the process, the counterion concentration of the AEX column load is not a controlled variable, but instead will show variability about a mean value; the management of the ProA column and the titration of the ProA peak will determine the normal variation about this mean. Three lots of AEX resin were enrolled in the study, comprising the minimum, maximum, and average exchange equivalents of the resin which could be provided by the vendor. This range of ion capacity was not achieved through manipulation of the resin manufacturing process, but rather reflects the range of ion capacities of resin lots that were available from the vendor's inventory.

For this WPC step, the column loading target is approximately 250 g/L, a rather high challenge, so product losses are minimal. The focus of the robustness study, therefore, was on the product quality of the AEX pool, and so the column load challenge was set at the worst-case conditions of 250 g/L for all runs. Product losses from these

Table III. Robustness study summary for a weak partitioning chromatography step.

pH	[Cl$^-$] (mM)	Resin ion capacity	HMW (%)	HCP (LRV)	Protein A (LRV)	Product loss (%)
8.2	18–28	Low and high	<0.7	>3.0	>3.2	<6
8.1	23	Medium	<0.7	>3.0	>3.2	<5
8.0	18	Low and high	<0.7	>3.0	>3.2	<4
8.0	28	Low and high	<0.7	>3.0	2.3, 3.0	<4

runs can then be used to estimate losses for smaller loads by proportionality (a 125 g/L load challenge would roughly double the percentage of product lost at 250 g/L loading for the same conditions, but only show similar or superior impurity clearance).

Table III shows the HCP and ProA residuals in the AEX pools. In all cases, the HCP content of the pool was below the limit of quantitation of the assay, and thus the HCP LRVs were >3 for all cases. The ProA levels of the AEX pool were also below the limit of quantitation for seven of the nine conditions. The worst-case for leached ProA removal is the lowest pH and highest counterion concentration, where both low- and high-density resins had measurable ProA residuals in the product pools. The ProA removal by the AEX step was still adequate under these conditions (2.3 and 3.0 LRVs, for the low- and high-density resins, respectively), resulting in ProA levels all less than 10 ppm, which is an acceptably low level of residual ProA (Jungbacker, 2004). The HMW levels were <1% for all conditions. The HMW in the load is only a few percent, and the overall step reduction of HMW is just two- to four-fold.

The robustness of several AEX-WPC steps have been tested for various mAbs using studies such as these. There was no evidence of significant changes in step performance over similar ranges of pH and counterion concentration.

Process Integration Into a mAb Purification Platform

AEX-WPC is well suited for use in a mAb purification platform. As mentioned above, the WPC stages of operation operations resemble those of the FT mode, and benefit from the simplicity of an isocratic chromatography step. The number of buffers, peak pooling criteria, and chromatographic control skid programming are very similar to a FT step.

Managing the interface between the ProA and an AEX-WPC step will minimize the variability of the composition of the AEX load. The ProA elution buffer should be compatible with AEX; glycine works well because it is a zwitterion at neutral or slightly basic pHs. Citrate or phosphate, conversely, will be counterions for the AEX step, which could cause complications in maintaining constant pH during operation. ProA pool titration enables control of the AEX load pH, provided the volume of titrant addition is easily controlled and the pool mixing is rapid. The typical control range of ±0.1 pH unit is not difficult to maintain for a well-designed titration step.

The counterion concentration in the AEX load can be controlled by formulating the ProA step pre-elution wash and elution buffers with the desired counterion concentration. If chloride is the only counterion in the AEX load, then the pre-elution wash buffer (typically a low concentration of Tris buffer plus sodium chloride) and elution buffer (glycine at pH 2.8–3.1 plus sodium chloride) should be formulated with the same chloride concentration as the WPC conditions targeted for the AEX step. This way, the pooling criteria for the ProA column will not influence the counterion concentration of the ProA pool, regardless of whether the ProA column is loaded to full capacity or not. (Highly loaded ProA columns have smaller pre-peaks, and include more of the pre-elution wash buffer in the product pool.) The low pH ProA pool is then titrated to the target AEX load pH using a concentrated buffer such as Tris base, which would only marginally dilute the load, and not change the counterion concentration. Thus, the main sources of variability in the counterion concentration arise from the formulation of the ProA pre-elution wash and elution solutions, which should be only a few percent at most, based on the process capability of the solution make-up equipment.

For a platform process used for production of several mAbs, the AEX WPC step will likely be run under unique conditions of pH and counterion concentration for each of

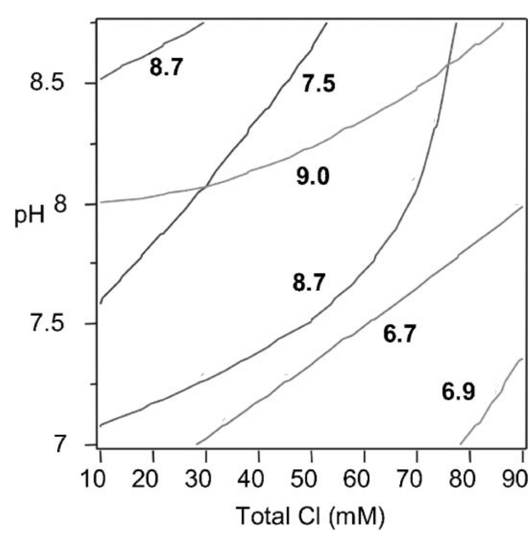

Figure 9. A panel of monoclonal IgGs display a wide range of binding strengths to the AEX resin. (The curves indicate the combination of pH and counterion concentration where the product K_p is equal to unity.) Note the differences in the product K_p position and slope for the various antibodies. The theoretical isoelectric point of each mAb is shown.

the different products. As shown in Figure 9, the contours of constant product K_p equal to unity range widely for six mAbs. There is no simple relationship between the shape or position of these contours and the theoretical protein p*I* (Bjellqvist et al., 1993) or amino acid composition. The position and slope of the curves varies from pH 7.0 to 8.75, for chloride concentrations from 10 to 90 mM. No single combination of pH and chloride concentration will result in a WPC platform for many different mAbs. While a routine optimization of AEX-WPC for each individual mAb requires some extra time relative to a mAb process platform that uses standardized AEX conditions for all mAbs, the cost is more than offset by the benefits of not requiring a third chromatographic step to complete the polishing of the ProA pool.

In some cases, solubility limits of the neutralized ProA pool will prevent WPC operation at low chloride concentrations (often driven by HCP solubility, not product solubility). These solubility limits can be easily established through the use of HTS studies designed to provide information on product K_p, if light scattering measurements are made of the neutralized ProA pool following adjustment to the appropriate pH and chloride concentration, prior to contacting this solution with the resin.

For very large scale production of mAbs, a single bioreactor harvest is commonly processed through several cycles of the ProA capture column. Because of the large loading capacity of the AEX-WPC step, it may not be necessary to cycle the AEX column. If the ProA column is loaded to 50 g/L and cycled five times, an AEX column of the same size as the ProA would process the entire 250 g/L load challenge in a single cycle. AEX-WPC steps are amenable to processing high concentration feedstreams; we have processed neutralized ProA pools of 25 g/L and not seen any decrease in performance compared to more dilute feedstreams of 5–10 g/L.

cGMP Performance in a mAb Purification Platform

An AEX-WPC step has been incorporated into the two-column platform purification process at Wyeth now for over 3 years. The average yield for all clinical cGMP AEX steps was 90%, which is an excellent average for clinical manufacturing operations not optimized to minimize process fluid losses to dead volumes such as extra column tubing and filter housings. The overall process yield of all the chromatography and filtration process steps following the bioreactor harvest is approximately 80%. The AEX-WPC steps provided significant reduction in HMW. Feedstreams with as much as 18% HMW were reduced to <2% in the AEX peak pool.

The impurity reduction by AEX-WPC was excellent, and the data for seven cGMP processes are summarized in Figure 10 (the data are reported as the average of all cGMP runs for each mAb, which range from 1 to 8 individual batches). Figure 10A–C shows the average HCP, ProA, and DNA levels for the ProA load (where appropriate), ProA pool, and AEX pool. In all cases, these impurities are reduced to low levels, well below targets commonly applied to cGMP mAb therapeutics in Phase I testing (Jungbacker, 2004). The impurity targets cited in this reference (100 ppm for HCP, 10 ppm for ProA, and 10 ng/dose for DNA) are included in the plots for context.

The virus clearance data for the two-column platform are summarized in Figure 10D. Three purification unit operations contribute to the viral safety profile of mAbs manufactured using this platform process. The ProA pool is held at low pH to provide virus inactivation. Viruses bind tightly to the AEX resin and are removed from the product stream. Finally, there is a virus removal filtration step. Figure 10D shows the clearance data for murine leukemia virus (MuLV), which is a model for the non-viable retrovirus-like particles present in CHO cell lines. All three virus clearance steps provide significant clearance of MuLV, with LRVs for each step being minimal values, as no viable virus was found after the pH hold (save for one study, which had detectable virus yet provided 6.1 logs of inactivation), in the AEX pool, or in the virus filter permeate. The total MuLV LRV for each process was greater than 15, which provided excess MuLV clearance of at least 5 logs in all cases (5 processes had total LRVs greater than 18, but for mAb #3, the virus stock had lower-than-expected titers, which depressed the LRVs for all steps).

The AEX-WPC step is an important contributor to viral clearance of the two-column platform process, as it is an orthogonal step to pH inactivation and virus filtration. WPC conditions are more stringent than FT operation, and therefore viruses should bind tighter, leading to high LRVs. It is important to consider the effect of the product K_p and load challenge on virus clearance, in order to avoid conditions that could cause virus impurity breakthrough due to competitive effects. In a virus clearance study which evaluated the effect of very long loadings in WPC operation, the virus broke through at the same time as all the other impurities such as ProA, HCP and DNA, consistent with a model of free competition between virus and other trace impurities for available binding sites on the resin. It is hoped that the data on virus clearance of the AEX step can be combined with additional process characterization studies to support a claim of modular or generic virus clearance (EMEA, 2006; Zoon, 1997).

Conclusions

An improved AEX step, utilizing the weak partitioning chromatography mode, is described for the purification of mAbs. In contrast to the standard flowthrough conditions used for AEX chromatography, a column loaded under WPC conditions results in a significant amount of product binding to the resin. These more stringent load conditions result in much stronger impurity binding, which improves the product pool purity. Stronger impurity binding also means that the column load can be greatly extended beyond standard FT operation, preserving the recovery of the step

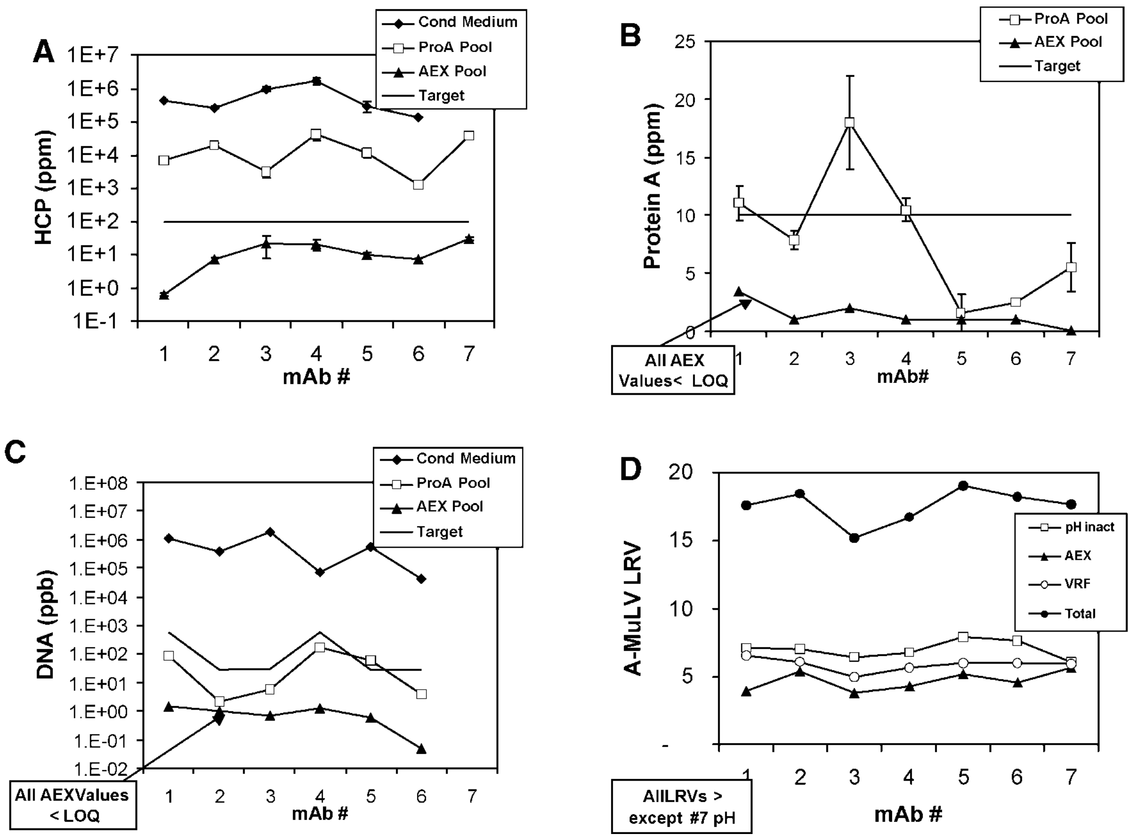

Figure 10. The performance of the AEX step for seven humanized monoclonal IgGs processes demonstrate the consistent and robust nature of the weak partitioning chromatography step. The average impurity levels of (**A**) HCP, (**B**) leached ProA, and (**C**) DNA are shown for cGMP production campaigns (operated at scales of 1–10 kg per batch). **D**: The MuLV virus clearance data are shown for the three steps contributing to viral safety for these platform processes (low pH inactivation, virus filtration, and weak partitioning AEX chromatography).

despite the increased product binding. A short isocratic wash after the load maximizes recovery, resulting in only a modest dilution of product load concentration. These improved performance parameters combined with greater loading are a significant advantage in very large-scale cGMP manufacturing of mAbs (Kelley, 2007).

Optimization of the AEX-WPC step requires tuning of the pH and counterion concentration for each mAb, which can be easily accomplished using high-throughput batch binding studies. The governing variable used for design of a WPC step is the product distribution coefficient, K_p, which is a fundamental descriptor of the strength of the protein–resin interaction. The improvement in impurity clearance can be correlated to the product K_p using either impurity LRVs or K_p. The product K_p is also used to define the operating window based on allowable ranges of product recovery and load dilution, and the minimum desired column loading.

The performance of the AEX-WPC step enables a two-column platform for cGMP purification of mAbs. Stringent product purity targets can be met without the need for a second polishing column following ProA capture. Seven consecutive clinical cGMP processes have used AEX in WPC mode, and consistently shown excellent clearance of host cell protein, leached ProA, DNA, and virus. High molecular weight species can also be reduced by WPC, in some cases from 18% down to <2%.

The principles of WPC have proven useful for other modes besides AEX (data not shown), and for proteins other than mAbs. HTS methods can be used to establish operating windows for other modes of chromatography (Kramarczyk et al., 2008; Wensel et al., 2008), and WPC may prove generally applicable to all adsorptive chromatography modes where the impurities to be removed bind more tightly to the resin than the product. As such, it is likely to provide a valuable alternative to bind-elute or standard flowthrough modes of chromatography.

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